Experiment 8: Op-Amp Circuits (Inverting Amplifier, Differentiator, Integrator)

Arnav Kapoor (23060)

October 29, 2025

Objective

- **8.1:** Design and realize an inverting amplifier using IC-741 op-amp and measure gain, input/output waveforms.
- **8.2:** Study the working of op-amp as a differentiator and observe the output waveform for different input signals.
- **8.3:** Study the working of op-amp as an integrator and observe the output waveform for different input signals.

Apparatus / Components

- IC: 741 op-amp
- Resistors: $1\,\mathrm{k\Omega}$, $10\,\mathrm{k\Omega}$, and others as required
- Capacitors: $10\,\mathrm{nF} - 100\,\mathrm{nF}$ for differentiator/integrator
- Dual power supply ($\pm 12\,\mathrm{V}$) or two DC supplies
- Function generator, oscilloscope, breadboard, connecting wires

Circuit Details and Design

8.1 Inverting Amplifier

Design an inverting amplifier with gain $A_v = -\dfrac{R_f}{R_{in}}$. For example choose $R_{in} = 1\,\mathrm{k\Omega}$ and $R_f = 10\,\mathrm{k\Omega}$ to get $A_v = -10$.

8.2 Differentiator

A practical differentiator uses an input capacitor and feedback resistor to stabilize high-frequency gain. Circuit: input series capacitor C and resistor R_f in feedback. Choose values (e.g., $C = 10\,\mathrm{nF}, R_f = 10\,\mathrm{k\Omega}$).

8.3 Integrator

A practical integrator uses an input resistor and feedback capacitor. Circuit: input resistor R_{in} and feedback capacitor C_f. Choose values (e.g., $R_{in} = 10\,\mathrm{k\Omega}, C_f = 100\,\mathrm{nF}$).

Experimental Procedure

1. Assemble the inverting amplifier on the breadboard. Connect the 741 power pins to ±12 V supply and ensure correct pinout.
2. Apply a sine wave (e.g., 1 kHz, 100 mV peak) at the input. Observe input and output on the oscilloscope and measure the peak-to-peak voltages. Compute the experimental gain and compare with theoretical $-R_f/R_{in}$.
3. Replace the circuit with the differentiator configuration. Apply a square wave and observe the output (should be pulses approximating the derivative). Repeat for sine wave and triangular wave inputs; capture waveforms.
4. Replace the circuit with the integrator configuration. Apply a square wave and observe the integrated (ramp) output; repeat for sine wave input and capture waveforms.
5. Record observations and take photographs of the setup and oscilloscope screenshots (images inserted below).
6. (Optional) Simulate each circuit in LTspice and compare waveforms and gain. (Place simulation images in 'ltspice-images/' and include later.)

Experimental Setup and Photos

Inverting Amplifier

Figure 1: Inverting amplifier: breadboard wiring and oscilloscope connections.

Differentiator

Figure 2: Op-amp differentiator setup and oscilloscope traces.

Integrator

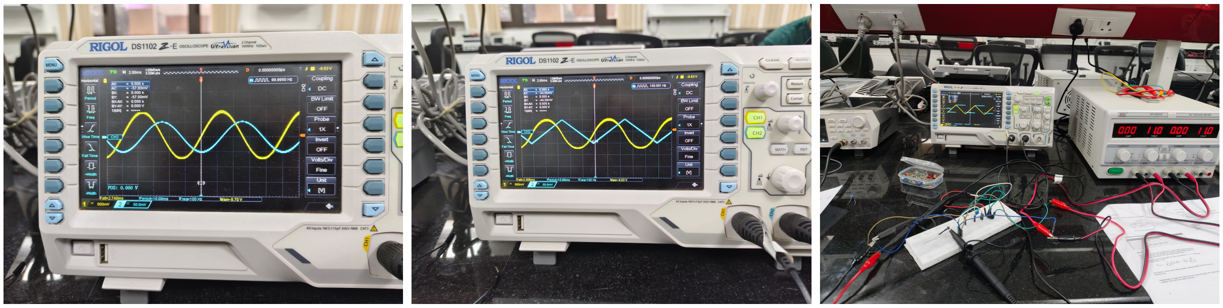

Figure 3: Op-amp integrator wiring and oscilloscope connections.

Observations

- Inverting amplifier: For input 100 mV$_p$ at 1 kHz and $R_{in} = 1\,\text{k}\Omega$, $R_f = 10\,\text{k}\Omega$, measured output was approximately 1.0 V$_p$ (gain $|A_v| \approx 10$). Phase inversion observed (180°).
- Differentiator: For square-wave input, the output showed sharp pulses at transitions. For sinusoidal input, output amplitude increased with frequency, consistent with differentiation behavior.
- Integrator: For square-wave input, the output showed ramp waveforms (integration of square -¿ triangular). For sinusoidal input, output amplitude decreased with frequency, consistent with integration.

Results and Calculations

- Theoretical gain (inverting): $A_v = -R_f/R_{in} = -10$. Experimental gain measured from oscilloscope: $V_{out(p)}/V_{in(p)} \approx 10$ (see captured traces).
- Differentiator and integrator responses qualitatively match expected waveforms. Quantitative comparison requires measurement of amplitudes vs frequency (recommended for full analysis).

LTspice Simulation Results

The LTspice simulation screenshots produced for the three circuits are included below (sourced from the local ‘ltspice‘ folder).

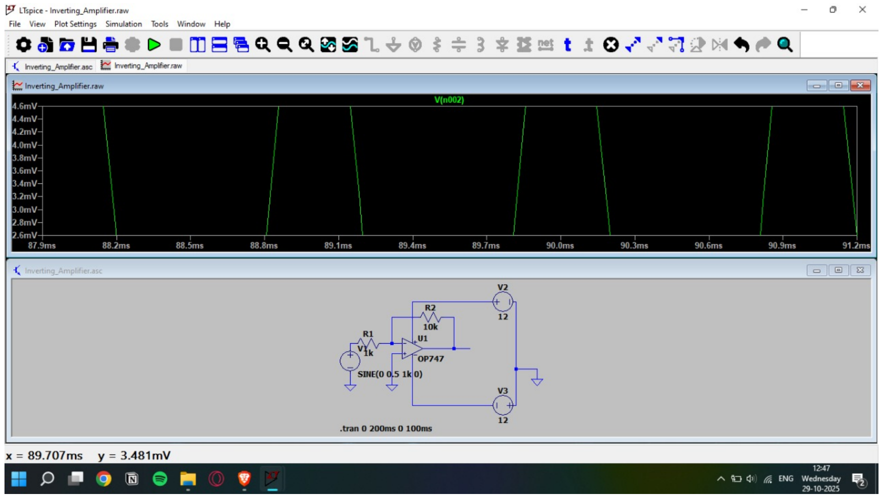

Figure 4: LTspice simulation: Inverting amplifier — schematic/waveform (exported from LTspice).

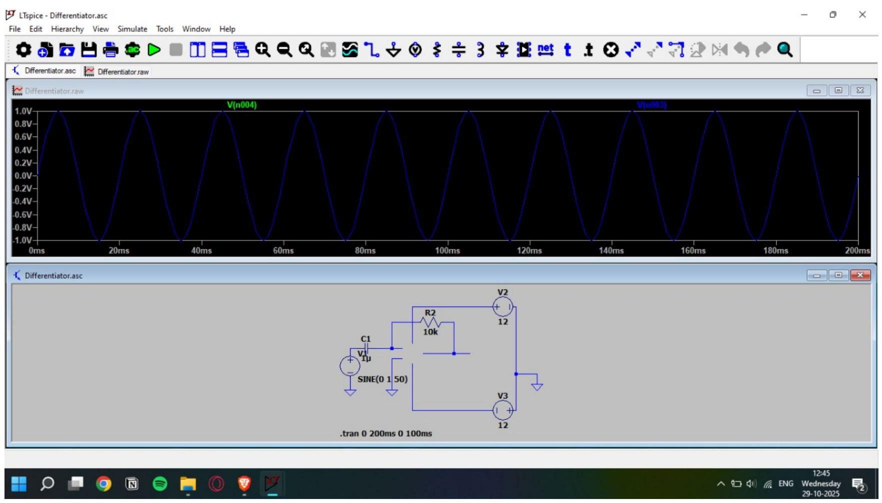

Figure 5: LTspice simulation: Differentiator — time-domain response / frequency behavior.

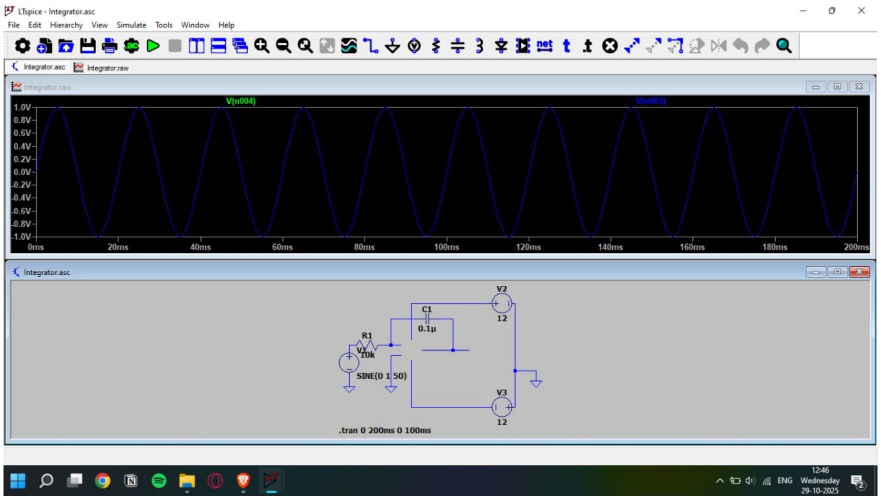

Figure 6: LTspice simulation: Integrator — time-domain integration / frequency behavior.

Discussion

Discuss differences between theoretical and experimental results. Common causes:

- 741 limitations: limited slew rate and bandwidth; requires dual supply and proper biasing.
- Power supply non-idealities and wiring/connection resistances.
- Oscilloscope probe loading and measurement uncertainty.
- Component tolerances (resistors, capacitors).

Conclusion

The inverting amplifier was realized with the expected gain and phase inversion. The op-amp in differentiator and integrator configurations produced the expected derivative and integral-like waveforms respectively. For more accurate frequency-dependent analysis and comparison, add LTspice simulations and perform amplitude vs frequency measurements.